INCH-POUND

MIL-M-28787/296A
AMENDMENT 4
16 December 1992
SUPERSEDING
AMENDMENT 3
13 January 1986

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, $\overline{JK}$, DIGITAL, KEY CODE GHK

This amendment forms a part of Military Specification MIL-M-28787/296A(EC), dated 14 July 1980, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, 7, 8, and 9 of this amendment.

PAGE 7

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, and F where applicable, followed by the remaining tests in the sequence."

4.5.1, fifth line: Delete "D" and substitute "E, G".

4.5.3, third line: Delete "D through" and substitute "E, G through".

4.5.3, sixth line: Delete "D" and substitute "E, and G".

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PAGE 8

6.5, second line: Delete "100°C" and substitute "85°C".

PAGE 9

6.5, Operating frequency, class I and class II: Delete "25" (two places) and substitute "18.2" (two places).

6.5, Output drive current, High level, class I and class II: Delete "-400 mA" (two places) and substitute "-400 max" (two places).

PAGE 17

FIGURE 7: Delete and substitute new figure 7 as printed on page 10 of this amendment.

PAGE 29

FIGURE 11: Delete and substitute new figure 11 as printed on page 11 of this amendment.

PAGE 31

TABLE IX, NOTE 2, first line: Delete "voltmeter V03, point C" and substitute "item M02 (point C)".

PAGE 37

FIGURE 12: Delete and substitute new figure 12 as printed on page 12 of this amendment.

PAGE 45

TABLE XII, Test conditions L1-L18, Initial limits, Min column: Delete "-0.1" (two places) and substitute "0.00" (two places).

PAGE 46

TABLE XII, Test conditions L19-L24, Initial limits, Min column: Delete "-0.1" (two places) and substitute "0.00" (two places).

PAGE 50

TABLE XIV, Test conditions N1 through N14, Min column: Delete "10" (two places) and substitute "5" (two places).

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TABLE XIV, Test conditions N15 and N16, Min column: Delete "20" (two places) and substitute "10" (two places).

PAGE 51

TABLE XIV, Test conditions N17 through N28, Min column: Delete "20" (two places) and substitute "10" (two places).

PAGE 69

*TABLE XVII, Test conditions W1-W14, Min column: Delete "110" (two places) and substitute "105" (two places).

PAGE 71

FIGURE 26, NOTE 1: Delete "PRP = 40 + 2.0, -0 ns (25.0 + 0, -1.2 MHz)" and substitute "PRP = 55 + 4.0, -0 ns (18.18 + 0, -1.23 MHz)".

PAGES 82 AND 83

TABLE XXI, Test conditions BB1 through BB24, Min column: Delete "5" (four places) and substitute "2" (four places).

PAGE 84

FIGURE 31, Note 3: Delete "Frequency = 120 Hz" and substitute "Frequency = 1 kHz".

FIGURE 31: Delete "Item C01" and substitute "Item C06".

FIGURE 31, Note 3: Delete "C01" and substitute "C06".

PAGE 88

FIGURE 33, NOTE 3a: Delete $t_{setup(1)}$ = 20 + 2, -0 ns" and substitute "$t_{setup(1)}$ = 30 + 3, -0 ns".

FIGURE 33, NOTE 3b: Delete "$t_{setup(0)}$ = 20 + 2, -0 ns" and substitute "$t_{setup(0)}$ = 30 + 3, -0 ns".

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FIGURE 35, NOTE 3b: Delete "$t_{hold(0)}$ = + 2, -0 ns" and substitute "$t_{hold(0)}$ = 5 + 2, -0 ns".

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B14	Vout (low-loaded)	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1,C2, C25,C26	Input breakdown current	V	S	0.0	200	0.0	200	0.0	200	0.0	200	0.0	200	μA
C3-C22, C27-C30	Input breakdown current	V	S	0.0	100	0.0	100	0.0	100	0.0	100	0.0	100	μA
C23,C24	Input breakdown current	V	S	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	mA
* D1-D14	Output short circuit current	VI	S	-6	-100	-6	-100	-6	-100	-6	-100	-6	-100	mA
E1-E14	Output leakage current	VII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
* F1-F24	Input clamp diode voltage	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G8	Functional test	IX	X	← Output must be as specified in table IX →										

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5° 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
H1-H14	Static output voltage, low level	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J14	Static output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K4	Input current, high level	XI	X	0	80	0	80	0	80	0	80	0	80	μA
K2,K3, K25,K26	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	μA
K5-K24, K27-K30	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	μA
L1,L4	Input current, low level	XII	X	0.00	-1.72	0.00	-1.72	0.00	-1.72	0.00	-1.72	0.00	-1.72	mA
L2,L3	Input current, low level	XII	X	0.00	-0.87	0.00	-0.87	0.00	-0.87	0.00	-0.87	0.00	-0.87	mA
L5-L24	Input current, low level	XII	X	0.00	-0.43	0.00	-0.43	0.00	-0.43	0.00	-0.43	0.00	-0.43	mA
* M	Power supply current	XIII	X	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	0.1	38.6	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N14	Transition time (tTLH), CL = 50 pF	XIV	S	5	50	5	50	5	55	5	50	5	60	ns
N15-N28	Transition time (tTLH), CL = 150 pF	XIV	Q	10	80	10	80	10	90	10	80	10	100	ns
P1-P14	Transition time (tTHL)	XIV	S	3	18	3	18	3	20	3	18	3	22	ns
P15-P28	Transition time (tTHL), CL = 150 pF	XIV	Q	6	40	6	40	6	42	6	40	6	44	ns
Q1-Q14	Propagation delay time from clock (tPLH), CL = 50 pF	XV	S	3	29	3	29	3	31	3	29	3	33	ns
Q15-Q28	Propagation delay time from clock (tPLH), CL = 150 pF	XV	Q	4	39	4	39	4	41	4	39	4	43	ns
R1-R14	Propagation delay time from clock (tPHL), CL = 50 pF	XV	S	4	44	4	44	4	46	4	44	4	48	ns
R15-R28	Propagation delay time from clock (tPHL), CL = 150 pF	XV	Q	5	54	5	54	5	56	5	54	5	58	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S8	Propagation delay time from set to Q (tPLH), CL = 50 pF	XVI	S	5	55	5	55	5	57	5	55	5	59	ns
S9-S16	Propagation delay time from set to Q (tPLH), CL = 150 pF	XVI	Q	5	65	5	65	5	67	5	65	5	69	ns
T1-T6	Propagation delay time from set to Q (tPHL), CL = 50 pF	XVI	S	6	65	6	65	6	67	6	65	6	71	ns
T7-T12	Propagation delay time from set to Q (tPHL), CL = 150 pF	XVI	Q	7	75	7	75	7	77	7	75	7	81	ns
U1-U6	Propagation delay time from clear to Q (tPLH), CL = 50 pF	XVI	S	5	50	5	50	5	52	5	50	5	56	ns
U7-U12	Propagation delay time from clear to Q (tPLH), CL = 150 pF	XVI	Q	6	60	6	60	6	62	6	60	6	66	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V8	Propagation delay time from clear to Q (tPHL), CL = 50 pF	XVI	S	6	63	6	63	6	67	6	63	6	71	ns
V9-V16	Propagation delay time from clear to Q (tPHL), CL = 150 pF	XVI	Q	7	73	7	73	7	77	7	73	7	81	ns
* W1-W14	Maximum operating frequency check, CL = 50 pF	XVII	S	105	120	105	120	105	120	105	120	105	120	ns
X1-X14	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Y1-Y14	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	μF
BB1	Input capacitance	XXI	Q	2	45	2	45	2	45	2	45	2	45	pF
BB2-BB4	Input capacitance	XXI	Q	2	30	2	30	2	30	2	30	2	30	pF

- See footnotes on next page.

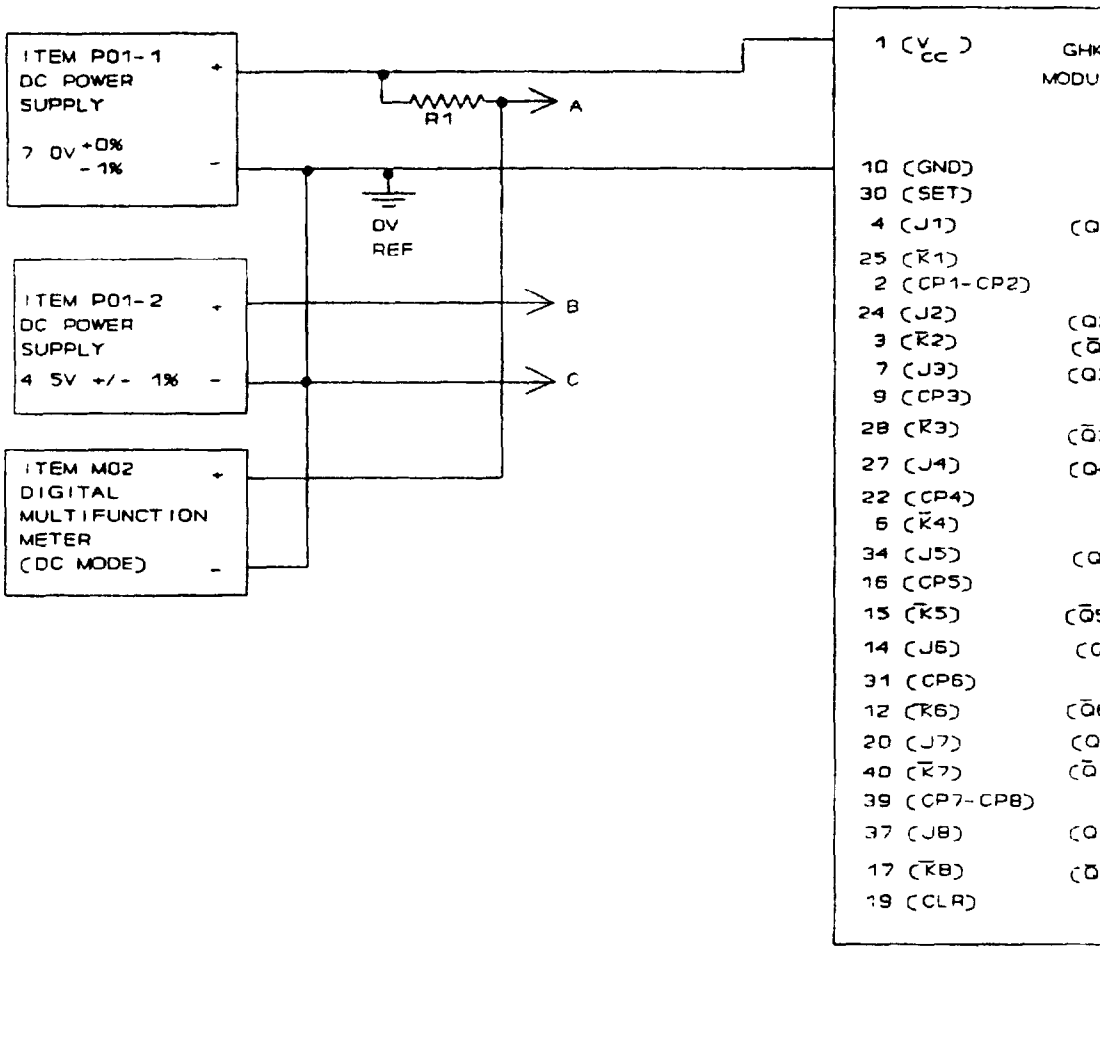
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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB5-BB24	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	2	15	pF
CC1-CC28	Setup time	XXII	S	170	230	170	230	170	230	170	230	170	230	ns
DD1-DD28	Hold time	XXIII	S	170	230	170	230	170	230	170	230	170	230	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

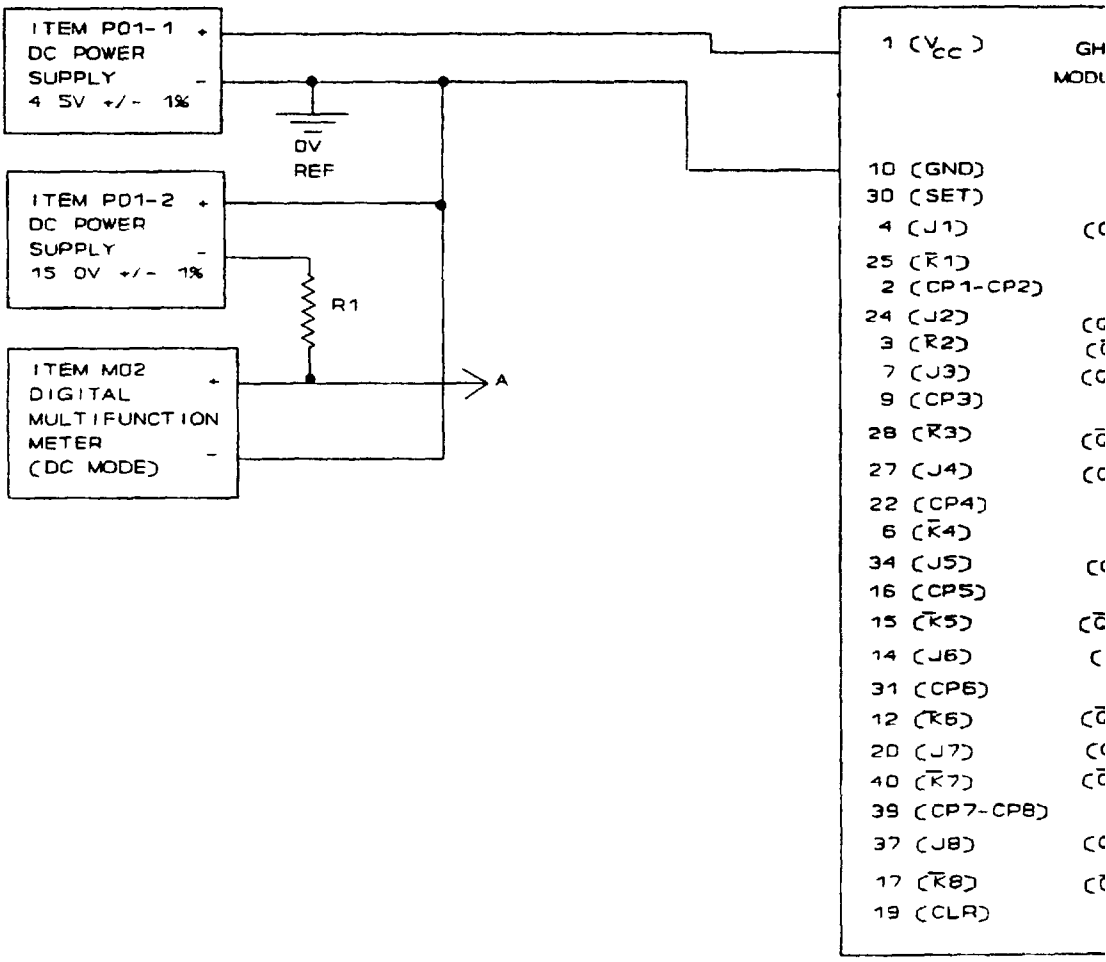
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NOTE: R1 = 1.27 kΩ ± 1%, 1/4 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

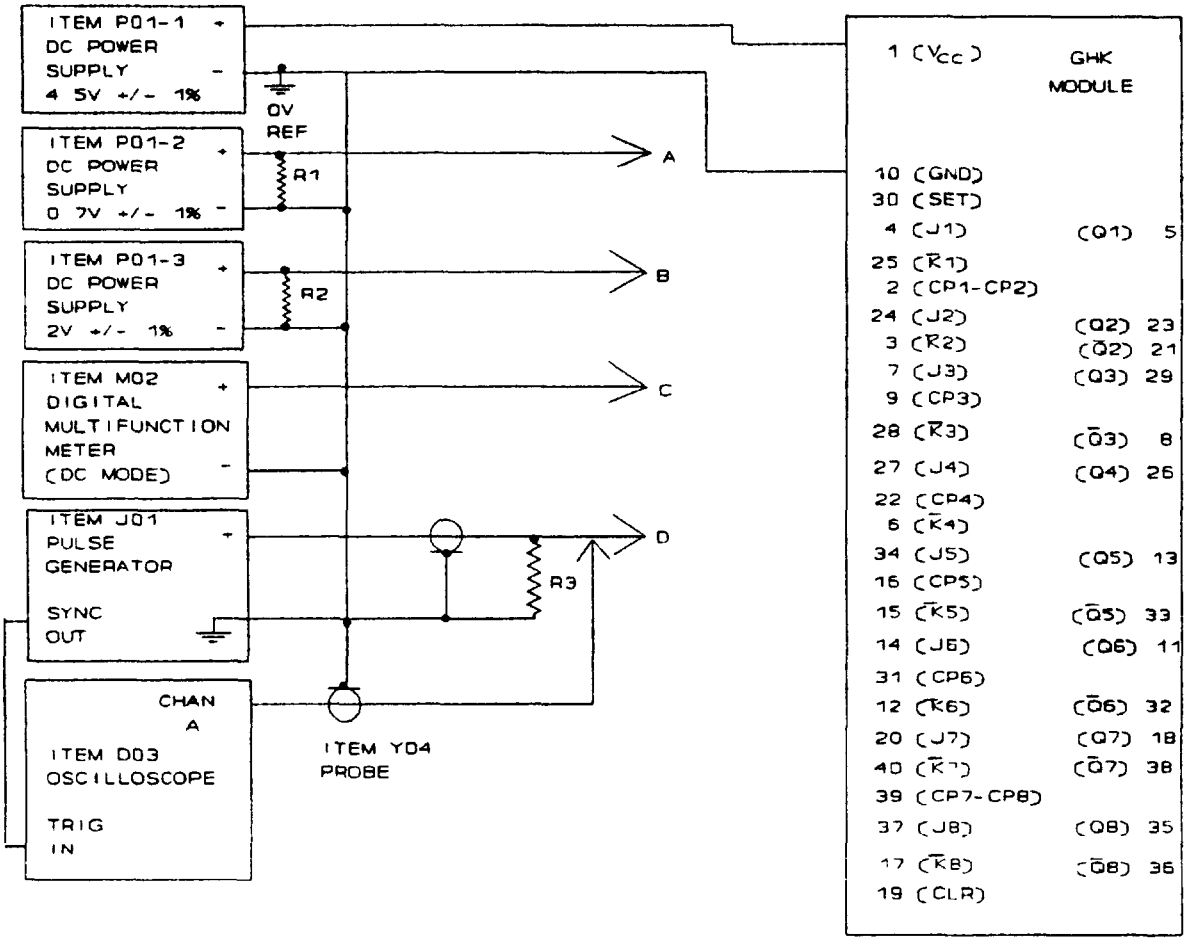
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NOTE: R1 = 750Ω ± 1%, 1/4 W.

FIGURE 11. Input clamp diode voltage test circuit for table VII

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NOTES:

1. R1 - R3 = 51Ω ± 5%, 1/2 W.
2. For test conditions G1 - G8 and J1 - J14, make all input circuit connections for a test condition, then use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
3. For test conditions H1 - H14 and J1- J14, load the output under test and the corresponding complement output (where available) with the load circuit per figure 14. Load circuits are not required for test conditions G1 - G8.

FIGURE 12. Functional and static output voltage (low-level and high-level) and input threshold check test circuit for tables IX and X.

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Navy - SH
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Navy - NW

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Review activities:
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